Multilevel Stress Detection with a Multimodal Approach: Combining Sensor Data

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***Abstract*—Stress significantly influences performance in daily activities, including academic tasks. This study aims to develop a method for acute stress measurement using a multimodal approach. We administered a C language exam and an oral arithmetic test to students while recording data with five different physiological sensors (EEG, ECG, EDA, EMG, and a respiratory sensor). The objective is to establish a multimodal method to assess stress and to analyze how different features derived from the collected data affect the accuracy of stress detection, recognizing the multidimensional nature of stress. The data were collected from 14 participants. Experimental results demonstrate that multidimensional data methods achieve better performance, the best accuracy was obtained with EDA and ECG, reaching 82% in a three-level stress classification.**

***Index Terms*—Electroencephalography (EEG), Electrocardiography (ECG), Electromyography (EMG), Electrodermal Activity (EDA), Visual Analog Scale (VAS), Lab Streaming Layer (LSL), Area under the curve (AUC), Sequential Feature Selection (SFS), Cross Validation (CV)**

I. Introduction

Stress is commonly defined as the physiological and psychological response to external demands, whether physical or mental. In an educational context, stress refers to students' reactions to academic and social pressures. Stress can have both negative and positive effects on academic performance, as well as on students' physical and mental health. It is also a contributing factor to conditions such as anxiety and depression. The literature identifies three types of stress: acute, chronic, and episodic. In this paper, we focus on developing a method to measure acute stress [1].

Physiological measures can serve as indicators of an individual's mental state. In this study, we administered a C programming exam to students with the aim of objectively assessing their stress levels and analyze the metrics of the physiological signals collected.

For physiological data collection, we used the PsychoKIT from Bitalino to record electrodermal activity (EDA), electrocardiogram (ECG), respiratory activity, and electromyogram (EMG) signals. Additionally, an OPENBCI electroencephalogram (EEG) system was employed. Figure 1 shows one participant during the experiment.

A comprehensive set of features is extracted from the collected data. This paper details the rationale behind the choice of sensors and the methods used for feature extraction, and discusses their relevance for stress detection. We initially recruited 3 first-year students from ECE engineering school who took a similar real exam one week after the data collection. To increase the number and diversity of students we also collected data from 11 interns from LIP6 and LyRIDS laboratories. They are fourth- and fifth-year students and had also previously taken C programming courses as part of their engineering studies.



Fig. 1. During experiment

II. Fundamental concepts and data collection protocol

A. Stress induction

A meta-analysis examining cortisol responses to acute mental stress identified several factors that elicit particularly intense stress reactions: social-evaluative threat, novelty, lack of control, and motivated performance [2]. Social-evaluative threat, which arises when an individual perceives their performance is being judged by others, has a strong influence on the stress response. Tasks involving social-evaluative threat typically challenge core aspects of personal identity (such as intelligence or competence), often extending far beyond an

individual's usual abilities. Novel stressors place demands on the individual for which no coping strategies have yet been developed, and many tasks become less stressful with practice. Unpredictability is designed to disrupt previously developed coping mechanisms (for example, by suddenly increasing the pace during mental arithmetic tasks). Performance may be motivated by the prospect of gaining a reward for success or losing something in case of failure. However, for performance to be motivating in these tasks, it must be measurable (for example, by counting errors in arithmetic tests).

Key elements include time pressure, negative feedback, social evaluation, and competition, all of which mirror the anxiety experienced during real-life examinations. This approach combined unpredictability (time limits) with social-evaluative threat (negative feedback on errors) and has proven to be an effective acute stressor. In the next subsection we present the elements that composed our data collection protocol.

B. Data collection

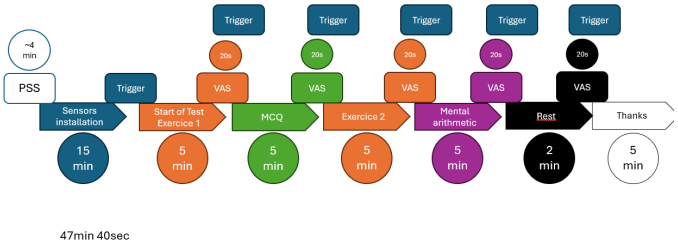


Fig. 2. Overview of the activities the students performed during data collection

1) Perceived Stress Scale: The PSS is a widely used and standardized questionnaire for specifically measures the perception of stress. It assesses how individuals interpret their stress levels, rather than directly measuring stress. The short version of Perceived Stress Scale (PSS) [12] was administred before the test to estimate the anxiety state and individual perception of stress. A total score on the PSS can be obtained by summing the scores of the 14 items (assuming the scale is unidimensional). A shortened 10-item version (with a score range from 0 to 40) can also be derived from the PSS items (items 1, 2, 3, 6, 7, 8, 9, 10, 11, and 14). This short version demonstrates the best discriminative sensitivity. [3] This questionnaire will serve as a reference point for the Visual Analog Scales (VAS), which subjectively measure the perceived level of stress, an experience that can vary from one individual to another. As for its original English version, one must first of all reverse the scores of the seven negative items (items 4, 5, 6, 7, 9, 10, 13).

2) C language exam: We administered a C language exam to participants to prepare the students for their actual exam, with the aim of inducing and measuring exam-related stress. The test content was fully aligned with the material covered in their real exam, which was scheduled for the following week. However, due to an insufficient number of participants, we included fourth- and fifth-year students who had previously studied C to increase the sample size.

3) Mental arithmetic: We included a five-minute oral mental arithmetic task, adapted from the Trier Social Stress Test (TSST) [13], to induce stress through cognitive load and social anxiety, with negative feedback given for errors. This task proved to be a strong stressor, as participants reported higher stress levels during this part compared to other parts. Additionally, the novelty is recognized as a known stress-inducing factor.

4) Rest: At the end of the test, participants were asked to rest for 2 minutes and to avoid any movements or talking to minimize artifacts.

5) Visual Analog Scale: Immediately after each induced stress or relaxation session, participants rated their perceived stress intensity on a scale from 0 (completely relaxed) to 100 (maximum tolerable stress).

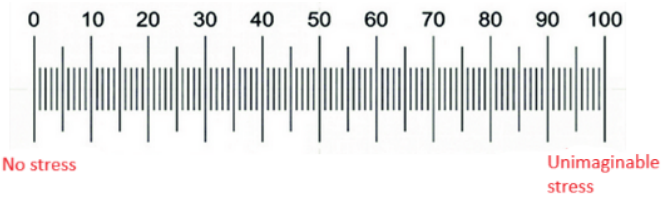


Fig. 3. Visual Analog Scale (VAS) used to register declarative stress during the exams

C. Data Analysis

After data collection was completed, the features were extracted from the raw physiological signals. The following subsections describe the preprocessing steps performed for each signal.

1) ECG: From the ECG data, we aim to compute heart rate variability (HRV) metrics, specifically the R–R interval, which is the time between successive R peaks. In order to clean the ECG signal, a 3th-order zero-phase Butterworth band-pass filter with cut-off frequencies of 5 and 15 Hz was applied. The main HRV features extracted are summarized in Table I. NeuroKit was used to compute HRV features [15], allowing us to extract most of the desired metrics. In addition, we retained the features available from the library, excepts the features that contains NaN or infinite values, which are listed below, :

```
HRV_DFA_alpha2, HRV_MFDFA_alpha2_Width,
↪ HRV_MFDFA_alpha2_Peak,
↪ HRV_MFDFA_alpha2_Mean,
↪ HRV_MFDFA_alpha2_Max,
↪ HRV_MFDFA_alpha2_Delta,
↪ HRV_MFDFA_alpha2_Asymmetry,
↪ HRV_MFDFA_alpha2_Fluctuation,
↪ HRV_MFDFA_alpha2_Increment, HRV_ULF,
↪ HRV_VLF, HRV_SDANN1, HRV_SDNNI1,
↪ HRV_SDANN2, HRV_SDNNI2, HRV_SDANN5,
↪ HRV_SDNNI5
```

The total number of features is 81. The most commonly

TABLE I
TIME-DOMAIN, NONLINEAR, AND FREQUENCY-DOMAIN CHARACTERISTICS OF RR INTERVALS REPORTED IN LITERATURE

No.	Description	Unit	Feature
Time-domain and statistical features			
1.	Mean of RR intervals	ms	mRR
2.	Standard deviation of RR intervals	ms	SDRR
3.	Standard deviation of normal-to-normal (NN) intervals	ms	SDNN
4.	Standard deviation of the averages of NN intervals calculated over 5-minute segments	ms	SDANN
5.	Mean of the standard deviations of NN intervals calculated over 5-minute segments	ms	SDNN Index
6.	Root mean square of successive differences between RR intervals	ms	RMSSD
7.	Mean of successive differences between RR intervals	ms	mSD
8.	Standard deviation of successive differences between RR intervals	ms	SDSD
9.	Coefficient of variation of RR intervals	%	CVRR
10.	Percentage of successive RR interval pairs that differ by more than 50 ms	%	PNN50
11.	Percentage of successive RR interval pairs that differ by more than 20 ms	%	PNN20
12.	Mean heart rate	Bpm	mHR
13.	Minimum RR interval value	ms	minRR
14.	Maximum RR interval value	ms	maxRR
15.	Mean absolute values of first differences of RR intervals	n.u.	δ RRI
16.	Mean absolute values of second differences of RR intervals	n.u.	γ RRI
17.	Interquartile range of RR intervals	n.u.	QD
18.	Integral of the RR interval density histogram divided by its height	n.u.	HRV triangular index
19.	Baseline width of the RR interval histogram	ms	TINN
Nonlinear (geometric) features			
20.	Cardiac sympathetic index	n.u.	CSI
21.	Cardiac vagal index	n.u.	CVI
22.	Standard deviation along the T direction (perpendicular to the line of identity) of the Poincaré plot	ms	SD1
23.	Standard deviation along the L direction (along the line of identity) of the Poincaré plot	ms	SD2
24.	Ratio between SD1 and SD2	%	SD1/SD2
25.	Area of the ellipse of the Poincaré plot, representing total HRV	ms^2	S
26.	Measures the regularity and complexity of a time series	n.u.	ApEn
27.	Measures the complexity and regularity of RR intervals	n.u.	SampEn
28.	Estimates the minimum number of variables required to model cardiac system dynamics	n.u.	D2
Frequency-domain (spectral) features			
29.	Total power of RR intervals in the ultra-low frequency band (0.003 Hz)	ms^2	ULF
30.	Total power of RR intervals in the very low frequency band (0.0033–0.04 Hz)	ms^2	VLF
31.	Total power of RR intervals in the low frequency band (0.04–0.15 Hz), related to sympathetic and parasympathetic activity	ms^2	LF
32.	Total power of RR intervals in the high frequency band (0.15–0.4 Hz), specific indicator of parasympathetic activity	ms^2	HF
33.	Frequency of the peak within the LF band	Hz	LF peak
34.	Frequency of the peak within the HF band	Hz	HF peak
35.	Power ratio between LF and HF bands	n.u.	LFHF
36.	Total power of RR intervals	ms^2	TP
37.	Normalized low frequency power	%	nLFP
38.	Normalized high frequency power	%	nHFP

Note: n.u. = normalized unit (dimensionless)

reported stress-related factor influencing HRV variables is low parasympathetic activity, characterized by a decrease in HF and an increase in components of LF [4]. Some studies have found that an increase in the LF/HF ratio is indicative of heightened sympathetic activity, which corresponds to stress [5], [6].

2) Electrodermal activity: EDA refers to the electrical conductance of skin, it increases with heightened activity of the sympathetic nervous system, making it a marker of physiological stress. This measurement is non-invasive and easy to collect, requiring only two electrodes placed on the palm of the hand. In this study, both the tonic component (skin conductance level, SCL) and the phasic peaks (skin conductance responses, SCR) were analyzed. The phasic component reflects immediate short-term responses, while the tonic component represents longer-term changes [22]. For each feature, statistical metrics were computed to reduce the data to a single row per signal, thereby condensing multiple rows into a single summary. This paper calculates the area under the curve for each peak scr into a feature named SCR AUC. The main extracted EDA features are summarized in Table II. NeuroKit was used to compute most of the desired features and process the signal [15]. We used the NeuroKit library's `eda_process()` function to decompose the EDA signals into phasic and tonic components and to calculate various features. In addition, we retained the features available from the library. We also had to calculate some features ourselves, such as the area under the curve of the SCR and the state of change of the SCL. For some participants, the signals reached the sensors' maximum measurable values, resulting in flat signals that contained no meaningful information. Such signals were excluded from further analysis. For a comprehensive list of features extracted by NeuroKit, readers may refer to the official documentation. The total number of features is 38.

TABLE II
EDA EXTRACTED FEATURES

No.	Feature	Statistic
1	Number of peaks	Sum
2	Amplitude	Min, Max, Mean, Std
3	Height	Min, Max, Mean, Std
4	Peak duration	Min, Max, Mean, Std
5	RiseTime	Min, Max, Mean, Std
6	RecoveryTime	Min, Max, Mean, Std
7	SCL signals	Min, Max, Mean, Std
8	Phasic signals	Min, Max, Mean, Std
9	SCR AUC	Min, Max, Mean, Std
10	Rate of change of SCL	Min, Max, Mean, Std

3) Respiratory sensors: The respiratory sensor is included to improve the accuracy of the electrocardiogram-based analysis, as respiratory frequency influences HRV measurements [7]. The primary features extracted from the respiratory data are the respiratory rate and respiratory rate variability features extracted from NeuroKit [15], except for those listed below that were NaN or infinite:

```
RRV_DFA_alpha2, RRV_MFDFA_alpha2_Width,
↪  RRV_MFDFA_alpha2_Peak,
↪  RRV_MFDFA_alpha2_Mean,
↪  RRV_MFDFA_alpha2_Max,
↪  RRV_MFDFA_alpha2_Delta,
↪  RRV_MFDFA_alpha2_Asymmetry,
↪  RRV_MFDFA_alpha2_Fluctuation,
↪  RRV_MFDFA_alpha2_Increment,
↪  RRV_DFA_alpha1,
↪  RRV_MFDFA_alpha1_Width,
↪  RRV_MFDFA_alpha1_Peak,
↪  RRV_MFDFA_alpha1_Mean,
↪  RRV_MFDFA_alpha1_Max,
↪  RRV_MFDFA_alpha1_Delta,
↪  RRV_MFDFA_alpha1_Asymmetry,
↪  RRV_MFDFA_alpha1_Fluctuation,
↪  RRV_MFDFA_alpha1_Increment, RRV_LFn,
↪  RRV_HFn, RRV_VLF, RRV_SampEn
```

The total number of features is 17. To assess signal quality, we attempted to calculate both the respiratory spectrum peak and the proportion of the spectrum used to estimate respiratory rate; however, the results were inconsistent, likely due to the complexity of these calculations. Consequently, we decided not to include these quality check features. Instead, we directly observed that the signal contained artifacts and noise, as indicated by outlier respiratory rates, sometimes above 20 or below 12 breaths per minute. We attempted to preprocess the signal using a fourth-order Butterworth bandpass filter (0.03–0.9 Hz) and resampled it to 4 Hz, but the resulting Fr values did not consistently fall within the expected 12–20 breath per minute range. Consequently, we utilized the `nk.rsp_clean()` function from NeuroKit to preprocess the signal and then computed Fr as the median value returned by the `nk.rsp_rate()` function. We also experimented with the `nk.rsp_process()` function, but this approach resulted with Fr values that were further from the expected range. However, the respiratory signal exhibited noise. We chose to retain the entire signal for simplicity as we needed to train a model using all features simultaneously. Further noise attenuation, for example using linear regression, could be explored in future work.

4) Electromyogram: EMG sensors were placed on both trapezius muscles, following evidence that this placement reliably indicates stress-related muscle tension [8]. Features quantifying muscle activation and tension are extracted from the EMG data. EMG data can also be utilized to reduce noise in signals from other sensors. Raw EMG data were filtered using a fourth-order zero-phase Butterworth high-pass filter (cut-off frequency: 30 Hz) to remove low-frequency artifacts, followed by a notch filter centered at 50 Hz (Q=30) to attenuate powerline interference. In this article, E denotes the processed EMG signal. A corresponds to the absolute value of E, which is subsequently low-pass filtered with a fourth-order zero-phase Butterworth filter at 6 Hz. The features extracted from both EMG data are listed in Table III.

TABLE III
DESCRIPTION OF EMG FEATURES AND THEIR CORRESPONDING FORMULAS.

Symbol	Unit	Feature description	Equation
RMSE	mV	Root Mean Square of Electromyogram	$RMSE = \sqrt{\frac{1}{N}} \sum_{i=1}^{N} E^2$
RMSA	mV	Root Mean Square of Activation	$RMSA = \sqrt{\frac{1}{N}} \sum_{i=1}^{N} A^2$
MAV	mV	Mean Absolute Value	$MAV = \sqrt{\frac{1}{N}} \lvert A \rvert$
Var	mV	Variance	$VAR = \sqrt{\frac{1}{N}} \sum_{i=1}^{N} (A - \bar{A})^2$
Energy	W	Energy	$ENERGY = \sum_{i=1}^{N} (\lvert A \rvert^2)$
MNF	Hz	Mean Frequency	$MNF = \frac{\sum_{j=1}^{M} f_j p_j}{\sum_{j=1}^{M} p_j}$
MDF	Hz	Median Frequency	$\sum_{j=1}^{MDF} p_j = \sum_{j=MDF}^{M} p_j = \frac{1}{2} \sum_{j=1}^{M} p_j$
ZC	—	Zero crossing	$\begin{cases} 1, \text{if } x_i \cdot x_{i+1} < 0 \text{ and } \lvert x_i \rvert > \varepsilon \\ \text{and } \lvert x(i+1) \rvert > \varepsilon, \\ 0, \text{else.} \end{cases}$

5) EEG: We used a 8 channels EEG, the OpenBCI model, which have dry electrode. The frontal cortex performs a key role [14]. We wanted to measure in priority Fp1, Fp2, F3, F4 because It has been shown that alpha power asymmetry and inter-hemispheric asymmetry indicate mental stress [5]. The alpha band power and beta band power were calculated from each electrode and for both hemispheres. The alpha asymmetry was also tested from left to right hemisphere for (F3, F4) and (Fp1, Fp2) electrodes. It has been reported that EEG activity is well correlated with mental stress in terms of reduction in alpha power [10] and an increase in beta power [11].

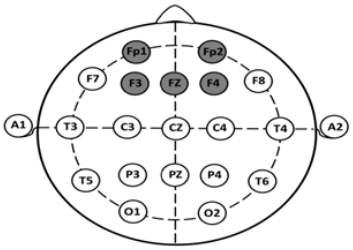


FIGURE 1. The 10-20 electrode placements.

Fig. 4. Channel placements

D. Hardware

To monitor and connect our EEG headset we used OpenBCI hardware, we applied a few setting and connected to the LSL the hardware. A fourth-order Butterworth bandpass filter, with a frequency range from 0.5 to 50 Hz, was applied to each channel. The signals were amplified with a gain of 6, and the headset operated via Bluetooth. To further attenuates blink and mouvement artefacts and reduce disproportionately the large delta-band power, we raised the low-frequency cutoff from 0.5 to 1 Hz by applying a 1 Hz high-pass filter. The second filter

TABLE IV
DESCRIPTION OF EEG FEATURES AND THEIR CORRESPONDING FORMULAS.

Symbol	Unit	Feature description	Equation
LAPF3	%	Normalized left hemisphere alpha band power (F3)	$\frac{\sum_{f=8}^{13} P_{F3}(f)}{P_{\text{tot},F3}} \times 100$
RAPF4	%	Normalized right hemisphere alpha band power (F4)	$\frac{\sum_{f=8}^{13} P_{F4}(f)}{P_{\text{tot},F4}} \times 100$
LAPFp1	%	Normalized left hemisphere alpha band power (Fp1)	$\frac{\sum_{f=8}^{13} P_{Fp1}(f)}{P_{\text{tot},Fp1}} \times 100$
RAPFp2	%	Normalized right hemisphere alpha band power (Fp2)	$\frac{\sum_{f=8}^{13} P_{Fp2}(f)}{P_{\text{tot},Fp2}} \times 100$
LBPF3	%	Normalized left hemisphere beta band power (F3)	$\frac{\sum_{f=13}^{30} P_{F3}(f)}{P_{\text{tot},F3}} \times 100$
RBPF4	%	Normalized right hemisphere beta band power (F4)	$\frac{\sum_{f=13}^{30} P_{F4}(f)}{P_{\text{tot},F4}} \times 100$
LBPFp1	%	Normalized left hemisphere beta band power (Fp1)	$\frac{\sum_{f=13}^{30} P_{Fp1}(f)}{P_{\text{tot},Fp1}} \times 100$
RBPFp2	%	Normalized right hemisphere beta band power (Fp2)	$\frac{\sum_{f=13}^{30} P_{Fp2}(f)}{P_{\text{tot},Fp2}} \times 100$
APA1	—	Alpha band power asymmetry (F3, F4)	$\frac{RAPF4-LAPF3}{RAPF4+LAPF3}$
APA2	—	Alpha band power asymmetry (Fp1, Fp2)	$\frac{RAPFp2-LAPFp1}{RAPFp2+LAPFp1}$

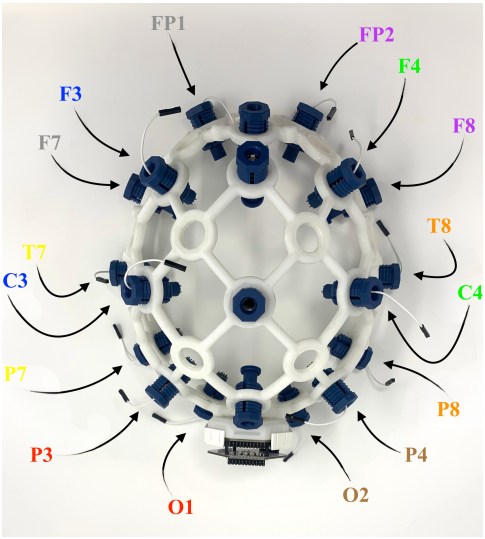


Fig. 5. Electrode placements on our headset

was also a fourth order filter, but since it was applied in zero-phase filtering, it resulted in an eighth-order filter. Combined with the first filter, the overall filtering process corresponded to a 12th-order filter. To monitor and connect EMG, ECG, EDA, and respiratory sensors, OpenSignals hardware was used in continuous mode and streamed data via the LSL protocol. This paper developed a PsychoPy interface allowing students to take the exam, alongside an HTML interface offering the same functionality. Both interfaces sent markers immediately after participants submitted their VAS responses. This approach proved highly effective for segmenting the signals corresponding to each part of the exam. Finally, all components were configured and synchronized using LabRecorder, which recorded the LSL stream and consolidated all data into a single XDF file with timestamps.

III. DATA PROCESSING

The data required preprocessing prior to analysis. Since the EEG headsets recorded at 250.25 Hz and the Bitalino system sampled at 1000 Hz, both signals were resampled to 250 Hz to allow concatenation using timestamps. Additionally, the event markers, along with participants' PSS and VAS scores, were concatenated with the physiological data. All the data was extracted from the XDF file except the PSS scores. Furthermore, we extracted all features from this processing and applied statistical methods to reduce the signals to a single row. During processing, signals were visualized to facilitate the identification of artifacts or issues. This approach helped explain the occurrence of unusual feature calculations. The signals were segmented using markers into five parts, corresponding to exercise 1, multiple choice questions, exercise 2, mental arithmetic and rest. This resulted in five rows of data for each participant. Before sending the data to machine learning algorithms, the data were separated into X (the features) and y (the label, which was the VAS score). 80% of the data was used for training, and 20% for evaluating the model on unseen data. The PSS score was always concatenated with the features for every signal. The features were scaled using a min-max scaling method. Stratified sampling was used to ensure that the distribution of stress levels remained consistent between the training and test sets. It is important to emphasize that the evaluation was performed on completely new data for the model.

IV. RESULTS AND DISCUSSIONS

EDA, EEG, ECG, Respiratory and EMG signals were analyzed and used to predict stress levels in participants.

The goal is to use all data from the sensors to perform a multidimensional analysis of perceived stress. Participants were asked to rate their stress on a scale from 0 to 100. However, due to insufficient data for a 101-class classifier, the stress levels were grouped into three categories: low (0 to 33.33), medium (33.33 to 66.66), and high (66.66 to 100). Unfortunately, Classes 1 and 2 had a distribution of 20 and 12.86%. They were underrepresented in the predicted scores due to the limited size of the dataset, and their precision was set to NaN for many folds. Their supports ranged between 2–3 and 1–2, respectively, while Class 0 had 9–10 supports, leading to better results for Class 0. The total support across all classes was 14 per fold. For every pipeline, a `RepeatedStratifiedKFold` cross-validation was performed with the same hyperparameters `n_repeats` = 5 and `n_splits` = 5 (25 folds). A `RandomizedSearchCV` was performed on the training data with a random forest classifier, utilizing an nested cross-validation strategy (inner and outer cv for Randomized search and model evaluation) based on `RepeatedStratifiedKFold`. Sometimes a `SequentialFeatureSelector` (SFS) from the `mlxtend` library was performed. When SFS is applied, the nested strategy is also used, including an inner CV for SFS, except for EDA and ECG signals. SFS was performed on the entire dataset rather than exclusively on the training set. This introduces an optimism bias, since, in practice, the test set should represent completely unseen data. However, we chose this approach because limiting SFS to only 80 % of the data would leave just 56 rows for selection, which could lead to unstable results. Each randomized search was conducted with 300 iterations and the following parameter grid: $n_estimators$: [50, 100, 200, 300]; max_depth: [None, 5, 10, 20, 30]; $min_samples_split$: [2, 3, 5, 10]; $min_samples_leaf$: [1, 2, 4, 6]; $max_features$: ['sqrt', 'log2', None]. We choose only one classifier to enable further analysis under computational constraints and aim to maintain similar conditions in order to compare results across different signals. All machine learning pipelines were implemented using `scikit-learn`, ensuring that each call to cross-validation, randomized search, or sequential feature selection (SFS) reinitializes the model parameters. This procedure reduces bias associated with multiple trainings and ensures robust model evaluation.

A. EDA analysis

The data was trained on several classifiers, including Gaussian Naive Bayes, Gradient Boosting, Random Forest, Support Vector Machine, Multilayer Perceptron, and K-Nearest Neighbors. The data was simply split in train and test. The results are show in Table V. A pipeline combining SFS and randomized search was executed. For computational efficiency, we first performed SFS on a reduced Random Forest model:

```
modele = RandomForestClassifier(
    n_estimators=50,
    max_depth=12,
    n_jobs=-1,
    random_state=0
)
```

Model	Accuracy
Gaussian Naive Bayes	58
Gradient Boosting	50
Random Forest	91.6
Support Vector Machine	83
Multilayer Perceptron	75
K-Nearest Neighbors	75

TABLE V
EDA MODEL ACCURACIES AS PERCENTAGES

A nested cross-validation strategy was employed, using RepeatedStratifiedKFold for the inner CV (RandomizedSearch) and the outer CV (Model evaluation). To limit the computational cost of SFS, we used a simple five-fold CV, and constrained the number of features to select (k_features) to the range (1,13). The choice not to embed an inner CV in SFS for this signal was due to the large number of features, which leads to high computational costs. After completing SFS, we conducted a RandomizedSearchCV with the selected features, followed by cross-validation and final testing on the test set. Executing a randomized search with the selected features is not optimal, because SFS should be integrated into the randomized search. This means that SFS would need to be re-executed for each iteration of the randomized search, resulting in an explosion of computational cost. The best hyperparameters identified were:

```
'n_estimators': 200, 'min_samples_split':
↪  10, 'min_samples_leaf': 6,
↪  'max_features': 'sqrt', 'max_depth':
↪  20
```

The selected features are listed below:

```
'RiseTime_std', 'RecoveryTime',
↪  'RecoveryTime_max', 'scr_auc'
```

The results are in Table VI. In Figure 6 the left panel shows the mean VAS scores for each part of the exam. Arithmetic was rated as the most stressful segment, which makes sense since it is part of the Trier Social Stress Test (TSST), a protocol specifically designed to induce stress. The right panel presents the average number of electrodermal peaks per participant in each segment. Electrodermal responses, which reflect activation of the sympathetic nervous system, occurred more often

Metric	Accuracy
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.66 ± 0.047
Cross validation	0.65 ± 0.053
ROC_AUC	0.60 ± 0.13
macro Precision	0.66 ± 0.047
macro Recall	0.33 ± 0.00
macro F1 score	0.27 ± 0.011
weighted Precision	0.66 ± 0.047
weighted Recall	0.66 ± 0.047
weighted F1 score	0.53 ± 0.059
avg total Support	12

TABLE VI
EDA PIPELINE RESULTS

during Arithmetic and Exercise 2 than during Rest. Figure 7 illustrates, on the left, the mean tonic skin conductance level (SCL) and the mean area under the curve (AUC) of all SCR peaks for each exercise and rest period. The SCL shows a modest increase during Exercise 2 and Arithmetic, but is not the lowest during the rest period, which is unexpected. The phasic SCR is an indirect measure of sympathetic autonomic activity. Here, the top three highest mean AUC values of SCR peaks correspond to Exercise 2, Exercice 1 and Arithmetic (by a small margin), while Rest shows the lowest values (also by a small margin). As arithmetic is known as a stressor, the AUC could reflect not only stress but also engagement. A different pattern is observed in students who are stressed but also more engaged. We identify two distinct response patterns when students are engaged in a coding exercise requiring more reasoning, something else occurs in the brain, a mix of stress and engagement. These different responses indicate that we can distinguish between two separate phenomena: stress and engagement. The interns had not learned how to complete Exercise 1 during their studies, which could explain why the mean AUC values of SCR peaks are lower than for Exercise 2.

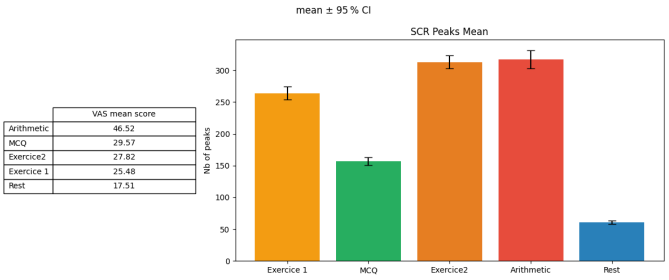


Fig. 6. Mean (± 95% CI) of VAS score of participants and Number(sum) of SCR peaks during all part of the experiment

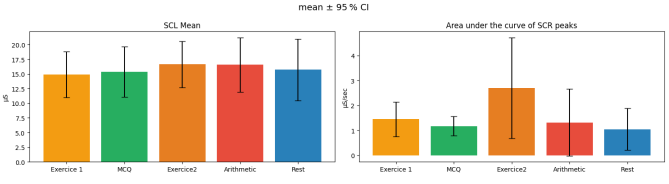


Fig. 7. Mean (± 95% CI) of tonic skin conductance level (SCL) and AUC of SCR peaks.

B. EEG analysis

Once the Randomized search was executed, the best hyperparameters identified were: `n_estimators = 300`, `min_samples_split = 3`, `min_samples_leaf = 4`, `max_features = None`, and `max_depth = 30`. An outer cross-validation was then performed, and the final model was evaluated on the test set, with results reported in Table VII. Then, we applied a SFS to perform an exploratory analysis using the same hyperparameters identified by the randomized search. The aim was to determine if accuracy could be further improved with the same model. The list of selected features are listed below:

Metric	Accuracy
Final Test accuracy (unseen data):	0.79
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.73 ± 0.056
Cross validation	0.72 ± 0.062
ROC_AUC	0.73 ± 0.1
macro Precision	0.71 ± 0.15
macro Recall	0.50 ± 0.13
macro F1 score	0.45 ± 0.13
weighted Precision	0.73 ± 0.08
weighted Recall	0.73 ± 0.06
weighted F1 score	0.64 ± 0.08
total Support	14

TABLE VII
EEG RANDOMIZED SEARCH RESULTS

```
['RAPF4', 'LBPF3', 'APAF3F4',
'Score_from_pss']
```

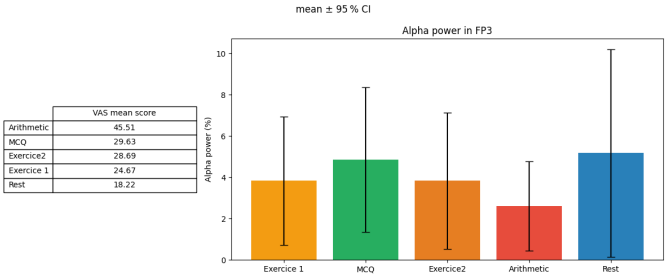
The results are shown in Table VIII. In Figure 8, decreases in alpha-band power are correlated with perceived stress, consistent with the findings of study [10]. On the other hand, Figure 9 shows that increases in beta-band power are not correlated with VAS scores, which contrasts with the findings of study [11]. The asymmetry may reflect a distinct, more prolonged form of stress, which could explain why the participant retains elevated stress levels during the rest phase. This is hypothetical, the observed asymmetry could be influenced by artefacts. The headset was not comfortable, and some participants, toward the end of the experiment, reported feeling pain from it.

Metric	Accuracy
Final test accuracy (unseen data):	0.79
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.76 ± 0.046
Cross validation	0.757 ± 0.045
ROC_AUC	0.69 ± 0.08
macro Precision	0.77 ± 0.13
macro Recall	0.59 ± 0.11
macro F1 score	0.53 ± 0.09
weighted Precision	0.77 ± 0.05
weighted Recall	0.76 ± 0.05
weighted F1 score	0.67 ± 0.05
total Support	14

TABLE VIII
EEG SFS RESULTS

C. ECG analysis

We first performed SFS on the same reduced Random Forest model as for EDA. To limit the computational cost of SFS, we used a simple three-fold CV, and constrained the number of features to select (k_features) to the range (5,25). The selected features are listed below:



	VAS mean score
Arithmetic	45.51
MCQ	29.63
Exercice2	28.69
Exercice 1	24.67
Rest	18.22

Fig. 8. Mean (± 95% CI)of VAS score of participants and Alpha band power during all part of the experiment

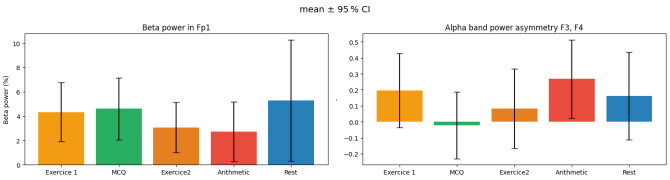


Fig. 9. Mean (± 95% CI) of Beta band power and F3, F4 asymmetry

```
'mSD', 'mHR', 'QD', 'gamma_RRI_nu',
 ↪   'LF_peak', 'HRV_pNN50', 'HRV_MinNN',
 ↪   'HRV_HTI', 'HRV_HF', 'HRV_LFHF',
 ↪   'HRV_LFn', 'HRV_LnHF', 'HRV_SD2',
 ↪   'HRV_IALS', 'HRV_PAS', 'HRV_SD2a',
 ↪   'HRV_SDNNa', 'HRV_MFDFA_alpha1_Max',
 ↪   'HRV_MFDFA_alpha1_Asymmetry',
 ↪   'Score_from_pss'
```

The best hyperparameters identified with Randomized search were:

```
{n_estimators: 100, min_samples_split: 2,
min_samples_leaf: 1, max_features: 'log2',
max_depth: None}
```

The results are in Table IX. A second manual feature selection

Metric	Accuracy
Final test accuracy (unseen data):	0.79
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.70 ± 0.094
Cross validation	0.71 ± 0.097
ROC_AUC	0.76 ± 0.11
macro Precision	0.66 ± 0.17
macro Recall	0.53 ± 0.17
macro F1 score	0.51 ± 0.18
weighted Precision	0.69 ± 0.12
weighted Recall	0.70 ± 0.094
weighted F1 score	0.65 ± 0.11
total Support	14

TABLE IX
ECG PIPELINE RESULTS

was carried out in an exploratory manner, based on simple criteria and inspired by the literature. The list of selected features is given below:

```
"HRV_MeanNN", "HRV_SDNN", "mSD",
 ↪   "HRV_CVNN", "HRV_pNN50", "HRV_pNN20",
 ↪   "mHR", "HRV_MinNN", "HRV_MaxNN",
 ↪   "HRV_SDRMSSD", "gamma_RRI_nu",
 ↪   "HRV_IQRNN", "HRV_HTI", "HRV_CSI",
 ↪   "HRV_SD1SD2", "HRV_S", "HRV_LF",
 ↪   "HRV_HF", "LF_peak", "HF_peak",
 ↪   "HRV_LFHF", "HRV_TP", "Score_from_pss"
```

A second SFS was executed on this list of features, yielding better results with only two features selected, using the same hyperparameters as before and without performing another randomized search.

```
'HRV_MinNN', 'Score_from_pss'
```

The results are presented in X. The final test accuracy without RepeatedStratifiedKFold is no longer shown because the accuracy is more stable and thus more representative with it.

Metric	Accuracy
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.74 ± 0.11
Cross validation	0.73 ± 0.12
ROC_AUC	0.84 ± 0.11
macro Precision	0.69 ± 0.16
macro Recall	0.66 ± 0.15
macro F1 score	0.63 ± 0.15
weighted Precision	0.77 ± 0.10
weighted Recall	0.74 ± 0.11
weighted F1 score	0.73 ± 0.10
total Support	14

TABLE X
ECG MANUAL SELECTION + SFS RESULTS

D. Respiratory analysis

We perform SFS, then Randomized search, and finally evaluate the model, using inner CV for both SFS and randomized search, and outer CV. The selected features are listed below:

```
'RRV_MedianBB', 'Score_from_pss'
```

The best hyperparameters identified were :

```
n_estimators: 50, min_samples_split: 2,
min_samples_leaf: 2,
max_features: None, 'max_depth': None}
```

Table XI shows the results.

Metric	Accuracy
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.67 ± 0.099
Cross validation	0.67 ± 0.099
ROC_AUC	0.80 ± 0.1
macro Precision	0.58 ± 0.18
macro Recall	0.52 ± 0.12
macro F1 score	0.51 ± 0.13
weighted Precision	0.66 ± 0.13
weighted Recall	0.67 ± 0.099
weighted F1 score	0.64 ± 0.1
total Support	14

TABLE XI
RESPIRATORY PIPELINE RESULTS

E. EMG analysis

Two sensors were used to collect EMG data from the right and left trapezius muscles. We performed three analyses: using the right, left, and both EMG signals as features for training. The feature ZC was dropped because, in every part of the experiment, the ZC per second was almost the same, making the feature not particularly relevant. Nevertheless, one machine learning pipeline was computed with it for the left EMG; the results showed 1% higher accuracy but 4% lower ROC_AUC, so it was not retained. A randomized search and model evaluation were conducted with RepeatedStratifiedKFold. Table XII shows the results for each case.

Metric	Accuracy
RepeatedStratifiedKFold (25 folds)	
Right EMG	
Accuracy	**0.66** ± 0.07
Cross validation	0.66 ± 0.071
ROC_AUC	0.71 ± 0.16
macro Precision	0.54 ± 0.21
macro Recall	0.39 ± 0.12
macro F1 score	0.34 ± 0.11
weighted Precision	0.65 ± 0.12
weighted Recall	0.66 ± 0.07
weighted F1 score	0.57 ± 0.08
Left EMG	
Accuracy	**0.68** ± 0.092
Cross validation	0.68 ± 0.091
ROC_AUC	0.72 ± 0.13
macro Precision	0.61 ± 0.22
macro Recall	0.49 ± 0.14
macro F1 score	0.47 ± 0.13
weighted Precision	0.66 ± 0.12
weighted Recall	0.68 ± 0.092
weighted F1 score	0.62 ± 0.078
Both EMG	
Accuracy	**0.64** ± 0.052
Cross validation	0.64 ± 0.051
ROC_AUC	0.68 ± 0.13
macro Precision	0.48 ± 0.18
macro Recall	0.32 ± 0.023
macro F1 score	0.26 ± 0.016
weighted Precision	0.60 ± 0.092
weighted Recall	0.64 ± 0.052
weighted F1 score	0.53 ± 0.054
total Support	14

TABLE XII
EMG RANDOMIZED SEARCH RESULTS

The accuracy of the left EMG is better, which contradicts the findings of study [16], where the right trapezius showed better results than the left. The lowest accuracy is obtained when using both trapezius signals, which may be due to the doubled number of features becoming too large compared to the small dataset.

F. PCA analysis

To enable comparison of the multimodality, a PCA was performed on each signal. This method has lower computational costs. SFS was also applied to some signal combinations. However, in some cases, it dropped all the features from one signal, making it difficult to compare the results based solely on accuracy. The features used were all those presented in the Data Analysis subsection for EDA, EEG, EMG, and Respiratory. For the ECG signals, three different groups of features were chosen because this signal contains too many features. The three groups are: all features, automatically selected features, and manually selected features (before SFS). This time, the data were not scaled before training. A pipeline was implemented as follows: selection of the feature group (only for the ECG signal alone), scaling, PCA, and a Random Forest Classifier. The pipeline was executed using a RepeatedStratifiedKFold. The parameter distribution is listed below:

```
"subset": [FeatureSubset(None),
 ↪  FeatureSubset(cols_A),
 ↪  FeatureSubset(cols_B)],
"pca__n_components": [0.70, 0.80, 0.85,
 ↪  0.90, 0.95, 0.99],
"rf__n_estimators": [50, 100, 200, 300],
"rf__max_depth": [None, 5, 10, 20, 30],
"rf__min_samples_split": [2, 3, 5, 10],
"rf__min_samples_leaf": [1, 2, 4, 6],
"rf__max_features": ["sqrt", "log2",
 ↪  None],
"rf__bootstrap": [True, False],
"rf__class_weight": [None, "balanced"]
```

The number of iterations in the randomized search was fixed at 300, except for ECG signals, where it was set to 900 due to the three groups of features in the parameter distribution. The results are presented in Table XIII. The results show that EDA and EEG combined gives worse results than alone. N is the number of rows in the dataset. The accuracy and balanced accuracy are increasing from 2 signals combined to 4 signals combined and decreases with all the signals combined. The main limitation of multimodality in this study is that EDA contains missing rows, which decreases the number of observations in the dataset.

Signals	Accuracy	ROC_AUC	B_Acc	N
RepeatedStratifiedKFold (25 folds)				
EDA	0.66 ± 0.075	0.65 ± 0.12	0.38 ± 0.11	70
EEG	0.68 ± 0.062	0.67 ± 0.10	0.38 ± 0.091	59
ECG	0.62 ± 0.12	**0.74** ± 0.076	**0.53** ± 0.15	70
Respiratory	0.67 ± 0.051	0.62 ± 0.14	0.36 ± 0.065	70
EMG	**0.69 ± 0.053**	0.68 ± 0.12	0.46 ± 0.11	70
EEG+EDA	0.62 ± 0.085	0.61 ± 0.13	0.35 ± 0.11	59
ECG+Respiratory	0.63 ± 0.11	0.59 ± 0.18	0.37 ± 0.093	70
EEG+EDA+ECG	0.65 ± 0.092	0.73 ± 0.13	0.48 ± 0.14	59
EEG+EDA+ECG+ Respiratory	**0.69** ± 0.094	0.67 ± 0.15	0.49 ± 0.14	59
EEG+EMG+ECG+ Respiratory	0.67 ± 0.092	**0.75** ± 0.086	0.45 ± 0.14	70
All signals	0.63 ± 0.11	**0.74** ± 0.12	0.42 ± 0.13	70

TABLE XIII
PCA PIPELINE RESULTS, B_ACC = BALANCED ACCURACY

G. EDA-EEG Combined Analysis

We executed the Pipeline SFS and Randomized search. The results are presented in Table XIV. The SFS results compared to PCA for the same combination are significantly better, providing strong accuracy with a small dataset.

Metric	Accuracy
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.77 ± 0.067
Balanced Accuracy	0.68 ± 0.094
Cross validation	0.77 ± 0.066
ROC_AUC	0.78 ± 0.10
macro Precision	0.74 ± 0.14
macro Recall	0.68 ± 0.094
macro F1 score	0.64 ± 0.12
weighted Precision	0.79 ± 0.098
weighted Recall	0.77 ± 0.067
weighted F1 score	0.74 ± 0.087
avg total Support	12

TABLE XIV
EDA-EEG SFS RESULTS

Work	Sensors	Nb	Period	Classes	fold	Acc
Ours	EDA, EEG, ECG, EMG, Respiratory	14	22 min	3 levels	25	82
[18]	ECG, PPG, GSR	17	50 min	2 levels	10	81.82
[8]	ECG, EMG	34	32 min	3 levels	-	96.7
[19]	IMU	59	15 min	2 levels	5	75
[20]	Empatica E4	15	hours	3 levels	1	91
[17]	BR, TI, Video, Audio, HR, SK	28	45 min	2 levels	LOSO(28)	89.4
[21]	PPG, GSR	32	-	2 levels	10	86.45

TABLE XVI
RELATED WORK, NB = NUMBER OF SUBJECTS, IMU = INERTIAL MEASUREMENT UNITS, HR = HEARTH RATE, BR = BREATH RATE, TI = THERMAL IMAGING, SK = SKIN CONDUCTANCE

H. Final results

Another SFS was performed on the combinations EEG, EDA, ECG and EEG, EDA, ECG and Respiratory. Only three features were chosen for the respiratory signal, given that it is noisy and limits the number of features. We selected Fr to see if it could help in combination with ECG and other signals. RRV_SD1SD2 indicates the balance between short-term and long-term variability of respiratory cycles.

```
'Fr', 'RRV_MedianBB', 'RRV_SD2SD1'
```

The automatically selected features were used for ECG. In both cases, SFS excluded the EEG and Resp signals and produced the same list of features. The parameters were optimized after the randomized search through cross-validation on the training data.

```
'n_estimators': 200, 'min_samples_split':
↪  3, 'min_samples_leaf': 1,
↪  'max_features': 'log2', 'max_depth': 5
```

Table XV presents the final results of this study.

Metric	Accuracy
RepeatedStratifiedKFold (25 folds)	
Accuracy	0.82 ± 0.094
Balanced Accuracy	0.76 ± 0.15
Cross validation	0.8024 ± 0.11
ROC_AUC	0.84 ± 0.12
macro Precision	0.82 ± 0.14
macro Recall	0.76 ± 0.15
macro F1 score	0.75 ± 0.15
weighted Precision	0.84 ± 0.10
weighted Recall	0.82 ± 0.094
weighted F1 score	0.80 ± 0.11
avg total Support	12

TABLE XV
FINAL SFS RESULTS

V. RELATED WORK

Our accuracy is slightly pessimistic compared to accuracies calculated with 10-fold cross-validation, making it more robust to randomness. However, using Leave-One-Subject-Out (LOSO) in study [17] provides an even more robust and realistic evaluation of machine learning models. The more stress levels (classes) are predicted, the lower the accuracy, even more with a small dataset.

VI. CONCLUSION

This paper aimed to identify a method for measuring stress in near real-time, using rapid techniques based on physiological measures as part of a multidimensional analysis of stress. The PCA and SFS methods show that multimodality tends to yield better accuracy and is a better approach for predicting stress, as stress has multidimensional causes and consequences. Although the results showed in Figure 6, 7, 8 and 9 generally align with expectations; indicating greater sympathetic activation during more stressful tasks; the metrics do not perfectly coincide across all measures. This partial consistency suggests that, while the data broadly reflect the relative stress levels of each condition, no single metric alone provides a definitive ranking of stress intensity. The PSS score was always retained by SFS and achieved the highest score among individual features for predicting the data, it is a relevant feature. It is important to note that the analytical model relies on a subjective stress scale rather than an objective measure. Subjective scores can vary widely between individuals experiencing the same physiological stress; the PSS attempts to normalize these variations across subjects. This creates a paradox for machine-learning: to train a model that accurately predicts stress, one would need a perfect stress-measurement tool as its ground truth. Should this model demonstrate robust performance with minimal invasiveness, it could be trained on a large and diverse population, potentially benefiting non-communicative people or with mental health disorders. We envision embedding it in an easy to use IoT device to make continuous stress monitoring accessible to everyone. Future work may combine other psychological signals and behavioral data to further reduce invasiveness. To overcome the need for ground truth labels during training, unsupervised learning methods could be explored to identify patterns of stress without relying on predefined annotations. Allowing it to be reused and adapted in different contexts (professional, school, sport stress) [17].

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